

LAB-1 : REQUIREMENTS ENGINEERING

NAME : HARSHITH R

SRN : PES1UG24AM117

SECTION : 5, 'B'

Requirements Table

Telemedicine Slot Booking & Prescription Portal

Actors: Patient, Attending Physician, Payment Gateway (external system).

Req ID	Type	Description	Priorit y	Acceptance Criteria	Rationale	Set
FR-001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.	High	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.	Ensures prescriptions are legally valid, tamper-evident, and traceable to a licensed physician.	Set 4
FR-002	Functional	The system shall allow a patient to search and filter available physicians by medical specialty and real-time slot availability.	High	Pass: Results show only physicians with ≥ 1 open slot in the next 7 days, correctly filtered by specialty. Fail: Results include unavailable or wrong-specialty physicians.	Core discovery function patients need before booking; reduces failed booking attempts.	Set 1

FR-003	Functional	The system shall allow a patient to book a specific video consultation slot, reserving it exclusively for that patient once payment is confirmed.	High	Pass: The booked slot disappears from all other patients' availability lists within 2 seconds. Fail: The same slot can be booked by two different patients.	Prevents doublebooking and guarantees the integrity of the appointment schedule.	Sets 1, 2
FR-004	Functional	The system shall automatically generate and transmit a unique, encrypted video-consultation room link to both patient and physician no later than 10 minutes before the scheduled appointment.	High	Pass: Both parties receive a working, single-use encrypted link ≥ 10 minutes before start time. Fail: Link is missing, reused across sessions, or delivered late.	Ensures secure, timely, and private access to the teleconsultation room.	Set 3
FR-005	Functional	The system shall allow a patient to download their digitally signed prescription as a PDF from their account within 24 hours of consultation completion.	Medium	Pass: Patient downloads a valid, signed PDF from the dashboard within 24 hours of consultation end. Fail: PDF is unavailable, unsigned, or late.	Gives patients timely access to prescriptions required for pharmacy fulfillment.	Set 4
NFR001	Nonfunctional (Performance & Security)	All video-session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load. Fail: Any data found unencrypted in transit or at rest.	Protects sensitive patient health data (PHI) and satisfies healthcare dataprotection obligations.	All sets
NFR002	Nonfunctional (Performance & Scalability)	The system shall support at least 500 concurrent video-consultation sessions with end-to-end video latency under 300 ms.	High	Pass: Load test at 500 concurrent sessions shows average latency < 300 ms with zero session drops. Fail: Latency exceeds 300 ms or sessions fail under peak load.	Guarantees usable, real-time video quality during peak clinic hours.	Set 3
NFR003	Nonfunctional (Availability & Reliability)	The system shall maintain at least 99.9% uptime during published clinic hours, with no confirmed booking or signed prescription lost during a failover.	High	Pass: 30-day monitoring shows ≤ 43 min downtime and a failover drill loses zero records. Fail: Uptime below 99.9% or any record lost.	Missed consultations and lost prescriptions carry direct clinical risk.	All sets

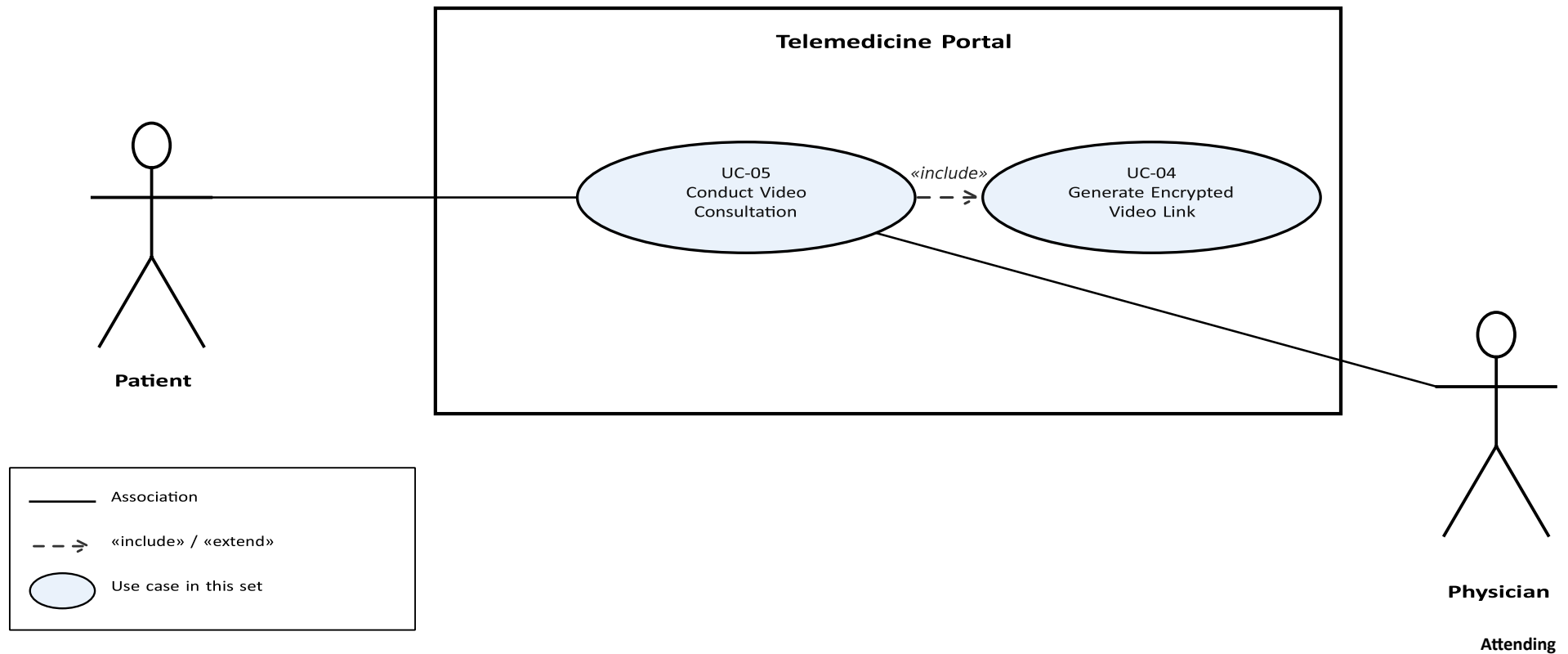
Module Split & Weightage

The eight use cases are divided into four equally weighted sets. Each set has its own use-case diagram and its own one-page use-case flow specification.

Set	Module Name	Use Cases Owned	Relationship Modelled	Weightage
Set 1	Doctor Discovery & Slot Booking	UC-01 Search Doctor by Specialty; UC-02 Book Consultation Slot	UC-02 «include» UC-03	25%
Set 2	Payment & Booking Changes	UC-03 Make Payment; UC-08 Cancel / Reschedule Slot	UC-02 «include» UC03; UC-08 «extend» UC-02	25%
Set 3	Tele-Consultation Session	UC-04 Generate Encrypted Video Link; UC-05 Conduct Video Consultation	UC-05 «include» UC-04	25%
Set 4	Prescription Lifecycle	UC-06 Author & Digitally Sign Prescription; UC-07 Download Prescription PDF	UC-06 «extend» UC-05	25%

Set 3 — Tele-Consultation Session

Weightage: 25% | Use cases: UC-04, UC-05



Use-Case Flow Specification

Telemedicine Slot Booking & Prescription Portal

Set 3 — Tele-Consultation Session | Weightage: 25%

Use Case: UC-05 — Conduct Video Consultation *(includes UC-04: Generate Encrypted Video Link)*

Primary Actor

Patient and Attending Physician

Preconditions

- A confirmed, paid booking exists for the current time slot.
- Both parties are authenticated and hold the single-use encrypted room link.
- Both endpoints meet the minimum bandwidth and grant camera/microphone permission.

Postconditions

- Success: The consultation is completed, session duration and clinical notes are stored encrypted, and the session is marked ready for prescription authoring.
- Failure: The session is marked incomplete, the reason is logged, and the case is queued for rescheduling or refund review.

Main Success Scenario

1. System generates and issues the single-use encrypted room link (UC-04) at least 10 minutes before the scheduled start.
2. Patient and physician each open the link; the system validates the token and the appointment window.
3. System admits the patient to a virtual waiting room and notifies the physician.
4. Physician admits the patient and the encrypted video session begins.
5. System records session start time and streams audio/video over the secured channel.
6. Physician records clinical notes in the consultation form during the session.
7. Either party ends the session; the system records the end time and total duration.
8. System saves the encrypted clinical notes and flags the consultation as complete, enabling prescription authoring (UC-06, Set 4).
9. Use case ends successfully.

Alternate Flow — 5a. Connection Drops Mid-Session

- a. 5a1. System detects loss of the media stream for one party and pauses the session timer.
- b. 5a2. The affected party is allowed to rejoin using the same single-use link for up to 10 minutes.
- c. 5a3. If neither party rejoins within 10 minutes, the system ends the session, marks it incomplete with the recorded reason, and queues it for reschedule or refund review.

